Image Copy Detection Using Dynamic Partial Function

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ABSTRACT

By extracting representative image features and employing our recently developed perceptual distance function (dynamic partial function), image copy detection can be performed effectively. Our empirical study shows that our scheme can detect various forms of near-replicas with high accuracy. Thus, our system has application for protection of copyrighted images and trademarks.

Keywords: copy detection, dynamic partial function, similarity

1. INTRODUCTION

Advances in the Internet and World-Wide Web technology have led to exceptional proliferation of information, disseminated at an unprecedented pace. However, the resulting widespread distribution of data (text, music, imagery and video) has increased the problems associated with copyright enforcement.

In this paper, we propose a copy-detection scheme to safeguard trademarks and copyrighted images. Given an image registered by its creator or distributor, our system checks to find out whether near-replicas of the image exist on the Internet and returns a list of suspect URLs. The creator or distributor of the image can then check the suspect images to determine whether they are indeed unlawful copies (for instance by using watermarks).

To achieve high detection accuracy, a system must address two central issues. First, representative features (signatures) must be extracted from each image. Second, effective distance function must be employed so that image similarity can be measured accurately. To address these issues, we use a set of multi-resolution features, including color, texture, and shape, for characterizing images. We employ dynamic partial function (DPF),[1] a distance function that we recently discovered through mining visual data, for measuring perceptual similarity. Our design goal is to cluster replicas together with their originals, and to keep them away from dissimilar images in the feature space. The results of our empirical study show that thanks to DPF our method works markedly better than our previous attempt.[2]

The rest of the paper is organized as follows: In Section 2, we explain how DPF was formulated, and why it works better than traditional distance functions. Section 3 describes our testbed and perceptual features. Section 4 reports experimental results. Finally, we offer our conclusion and discuss some future work in Section 5.

1.1. Related Work

Many watermark schemes have been proposed to safeguard copyrighted images. These schemes add to the images the creator's or the distributor's identity. Though useful, a watermark is vulnerable to image processing, geometric distortions, and subterfuge attacks.[2] In addition, working with a large variety of watermark schemes increases the detection time drastically since each Internet image may need to be processed many times, each time with a different watermark scheme.

Safeguarding copyright by checking the image directly has two distinct advantages. First, although a copied image can be altered, a pirate would be unlikely to change the image substantially and thereby lose its similarity to the original. Second, each image needs to be profiled only once. We believe that the watermark still plays an important role in authenticating images in courts of law. After the image copy system provides the creator or distributor with a suspect list, the rightful owner of the images can use watermark or other authentication techniques to prove the ownership.

The copy detection system may appear similar to some content-based image retrieval (CBIR) systems, such as QBIC,[3] Virage,[4] Simplicity,[5] etc., which retrieve images similar to a given one. We believe that being able to retrieve near-replicas reliably is the prerequisite of a CBIR system that can prove its effectiveness in finding semantically similar images. After all, if a system cannot even find replicas that are obviously similar, its effectiveness in achieving more ambitious goals could be questionable.

2. DYNAMIC PARTIAL FUNCTION

In this section, we start by examining the most popular Minkowski[6] distance function for measuring image similarity. We then present DPF and explain why it works effectively.

Suppose two images X and Y are represented by p dimensional feature vectors $(x_1, x_2, \ldots, x_p)$ and $(y_1, y_2, \ldots, y_p)$, respectively. We define the feature distance in feature channels i as $\Delta d_i = |x_i - y_i|$, for $i = 1, 2, \ldots, p$. We normalize feature values to be in the range of [0, 1]. Thus, the feature distance is also in the range of [0, 1].

The Minkowski metric is widely used for measuring image similarity. In this metric, the distance between images X and Y is defined as

$$D(X, Y) = \left(\sum_{i=1}^{p} \Delta {d_i}^r \right)^{\frac{1}{r}}. \tag{1}$$

When we set $r = 1$, the above distance is the City-block distance or L1 distance. When we set $r = 2$, it is Euclidean distance function or L2. In general, a Minkowski-like function considers all features in similarity measurement. However, the extensive study[1] shows that the above assumption is questionable.

Some important work in the cognitive psychology community has provided ample evidence that similarity is both dynamic and non-metric.[7] Most recently, Goldstone[8] shows that similarity perception is the process that determines the respects (i.e., features, or attributes) for measuring similarity. More precisely, a similarity function for measuring a pair of objects is formulated only after the objects are compared, not before the comparison is made; and the respects for the comparison are dynamically activated in this formulation process. Let us use a simple example to explain. Suppose we are asked to name two places that are similar to Los Angeles. Among several possibilities, San Francisco and New York could be two reasonable answers. However, the respects in which Los Angeles is similar to San Francisco differ from those in which Los Angeles is similar to New York. If we use the shared attributes of Los Angeles and San Francisco to compare Los Angeles and New York, the latter pair might not seem similar, and vice versa. Thus, a distance function ought to activate partial respects dynamically in comparing different pairs.

These observations have led to the design of the dynamic partial distance function. We define sets

$$\Delta d_i = \text{The smallest } m \ \Delta d\text{'s of } (\Delta d_1, \Delta d_2, \ldots, \Delta d_p)$$

Then the dynamic partial distance function is defined as

$$D(X, Y) = \left(\sum_{i \in \Delta_m} \Delta {d_i}^r \right)^{\frac{1}{r}}, \tag{2}$$

where m and r are two tunable parameters. When $m = p$, DPF degenerates to the Minkowski metric. When $m < p$, it counts the smallest m feature distances between two images, and ignores the $(p - m)$ largest feature distances, which may introduce noise. When a proper m value is chosen, the distance function clusters similar images more compactly and keeps dissimilar images away. By training a large image dataset, we could find an "on average optimal" m value. By using this m value, we could effectively find images that are perceptually similar.

The above formulation shows that DPF activates different features for different image pairs. The activated features are those with minimum differences—those that provide coherence between the images. If coherence can be maintained (because a sufficient number of features are similar), then the image pairs are perceived as similar.

3. IMAGE TESTBED AND PERCEPTUAL FEATURES

In this section, we describe how our image dataset was constructed in two parts:

1. Testbed setup (Section 3.1)
2. Low level feature extraction (Section 3.2)

3.1. Testbed Setup

To ensure that sound inferences can be drawn from our extensive experimental results, we carefully constructed the testbed.

First, in order to test the general performance of our methods, we employ a machine learning approach, which needs both training and independent testing datasets. The images in the testbed are collected from Corel image CDs and the Internet. We normalize these images into two sizes, either 384×256 or 256×384.

The training set consists of 116k images that are produced from 2k seed images. For each seed image, we conducted 57 different transformations, (which we will describe shortly), and hence form 2k similar image sets. These 116k images are used for discovering the best parameter settings of our methods, so that each similar image set (of 58 images) can be clustered very compactly in the feature space, and simultaneously dissimilar images can be kept away.

The independently constructed testing set includes 16k images. Among these, we identify 100 images as query images. For each query image, the same 57 transformations are conducted. These 16k images are used to examine the effectiveness of DPF in finding similar images.

Second, since an image can be replicated in different forms, to make our testbed comprehensive enough to cover reasonable and diversified transformations, we conduct the following 57 transformations:

1. Colorizing. Each image is colorized through red/green/blue channels by five percentages: 0/0/30%, 0/30%/0, 30%/0/0, 30/70%/90% and 50%/30%/0.
2. Modulation.For each image, we vary the amplitude of the hue/saturation/brightness channels by fifteen percentages: 40%/0/0, 60%/0/0, 80%/0/0, 120%/0/0, 140%/0/0, 0/40%/0, 0/60%/0, 0/80%/0, 0/120%/0, 0/140%/0, 0/0/40%, 0/0/60%, 0/0/80%, 0/0/120% and 0/0/140%.
3. Cropping. We symmetrically remove the outer borders to reduce each image by 5%, 10%, 20%, 30%, 40%, 50%, 60%, 70% and 80%, and then scale it back up to the original size.
4. Rotation. We rotate each image by 90, 180 and 270 degrees.
5. Scaling.
   - Scale up then down. Each image is scaled up by 5 and 10 times, and then scaled back to the original size.
   - Scale down then up. Each image is scaled down by factors of 2, 5 and 10, then scaled back to the original size.
6. Downsampling. We downsample each image by seven different percentages: 10%, 20%, 30%, 40%, 50%, 70% and 90%.
7. Shading. We shade the image using a distant light source, the position of which is specified by the two parameters, azimuth=1000, elevation=50.
8. Blurring. Each image is blurred by five Gaussian operators, of which the radius/standard deviations based on pixels are: 6/2, 6/3, 6/4, 12/4 and 12/8.
9. Framing. For each image, we produce five framed images, each with a randomly selected frame color. The frame around each image is 8 pixels in width.

10. Changing format. We obtain the GIF version of each JPEG image.
11. Changing contrast. We enhance the intensity differences between the lighter and darker elements of each image.

3.2. Feature Extraction

To describe images corresponding to the human vision system, we must find a set of features to follow human perception, which, as some researchers[8] have suggested, works in a multi-resolution fashion. In our study, we employ a multi-resolution image representation scheme and represent each image as a 240-dimensional feature vector, which includes three types of perceptual features: color, texture and shape. Based on the experimental results in Section 4, we found that by using texture features obtained from both Gabor filter bank and Wavelet filters, the system can reach higher accuracy than by using one of them alone. So specifically, to characterize each image, we use 104 global color features (Section 3.2.1), 36 global texture features from Wavelet filters and 96 global texture features from Gabor filters (Section 3.2.2). The shape features are involved in the color and texture features. Then, to ensure the equal emphasis of the feature components within a feature vector, we normalize each feature component from zero to one by using Gasussian normalization according to the 3-σ rule.[9]

3.2.1. Color

Color is one of the most important elements for human vision.[10] To capture the perceptual properties of images, it is desirable to use a perceptual color space, in which small changes in the color channels will correspond to small perceptual differences. The RGB color space does not have this property. But the HSV color space offers some perceptual continuity and is thus employed in our methods. Furthermore, research shows that, throughout the visible spectrum (roughly 380nm to 780nm), the colors that can be named by all cultures are generally limited to eleven,[8] namely *black, white, red, green, blue, brown, yellow, pink, purple, gray* and *orange.*

Based on the above research, we use 108 features to represent colors. We divide HSV space into twelve bins, including eleven culture color bins and one outlier bin. Within each bin, we extract features in the following way. At the coarsest resolution, we characterize color using color histogram. At finer resolutions, we record eight features for each color. These eight features are color means (in *H, S* and *V* channels), color variance (in *H, S* and *V* channels) and two shape characteristics: elongation and spreadness.[11] Color elongation characterizes the shape of a color, and color spreadness characterizes how that color scatters within the image.

3.2.2. Texture

Texture is another important element for human vision. As shown by many researchers,[10,12–15] characterizing texture features in terms of structuredness, orientation, and scale (coarseness) fits well with models of human perception. The texture features are extracted using both Wavelet and Gabor filters.

We choose a discrete wavelet transformation (DWT) using quadrature mirror filters because of its computational efficiency.[14] Each wavelet decomposition on a 2-D image yields four subimages: a $(\frac{1}{2} \times \frac{1}{2})$ scaled-down image of the input image and its wavelets in three orientations: horizontal, vertical, and diagonal. Decomposing the scaled-down image further, we obtain the tree-structured or wavelet packet decomposition. The wavelet image decomposition provides a representation that is easy to interpret. Every subimage contains information of a specific scale and orientation and also retains spatial information. We obtain texture features from subimages of three scales and three orientations. At each scale and each orientation, we extract four global texture features: energy mean, energy variance, elongation, and spreadness.

The Gabor representation has been shown to be optimal in the sense of minimizing the joint two-dimensional uncertainty in space and frequency. It is widely used for scale tunable edge and line (bar) detectors. Considering the computational efficiency, a bank of real, circularly symmetric Gabor filters[13] is employed. We use the filter bank with four scales, and the orientation of the filters changes by the step of $\pi/6$. The Gabor kernel sizes are 5×5, 9×9, 17×17, and 35×35. In each scale and each orientation, we use the same texture features: energy mean, energy variance, spreadness, and elongation.

4. EMPIRICAL STUDY

In our empirical study, we first examined three feature sets. We then evaluated the effectiveness of DPF on image copy detection using the most effective feature set.

4.1. Feature Selection

To find a robust set of image features for conducting copy detection, we employed DPF to examine three feature combinations using the 116K training dataset. We used DPF to find the the 57 near-replicas the original images. An effective feature set should return most of the 57 near-replicas in the top-57 query results.

Figures 1 (a), (b), and (c) show that the error rates of the three feature combinations.

- For color and Wavelet texture features, the lowest missing rate is 25.2%, where the best m value is $m = 94$.
- For color and Gabor texture features, the lowest missing rate is 22.5%, where the best m value is $m = 114$.
- For color, Wavelet and Gabor texture features, the lowest missing rate is 20.1%, where the best m value is $m = 120$.

Since the color, Wavelet and Gabor combination achieves the lowest missing rate, we used this feature set in the rest of our empirical study.

4.2. DPF

To test the effectiveness of DPF on image copy detection, our empirical study consists of two parts:

1. Using original images as queries, and
2. Using cropped and framed images as queries.

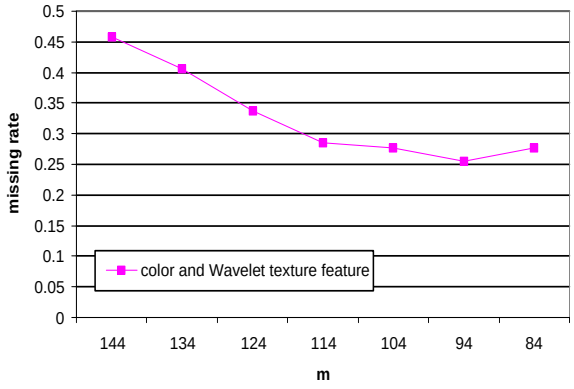
In each part, we first used the training dataset to predict the optimal m value. After setting DPF with the optimal m value, we used the testing dataset to examine DPF's effectiveness. We compared the precision/recall of the DPF, L2, and L1 distance functions. For all experiments, we set $r = 2$ for DPF.

4.2.1. Using Original Images as Queries

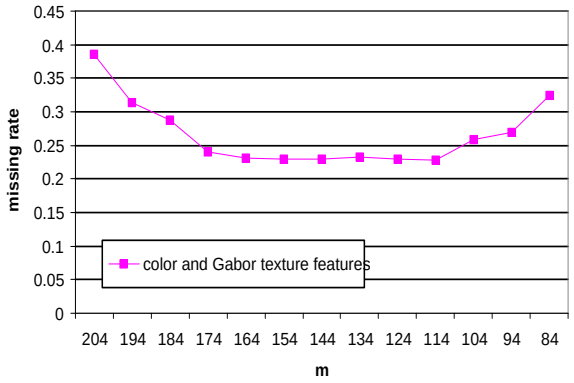
The most common copy-detection scenario is searching for near-replicas of original images. This is what we did in the first test. Our test is composed of two steps: training and testing.

Predicting parameter m

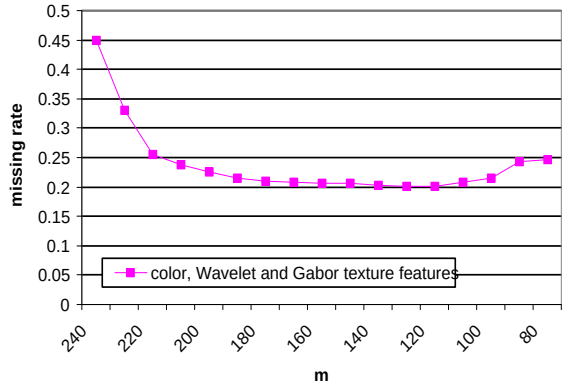
In order to find the "on average optimal" m value, we trained DPF using our 116K training image dataset. We used the 2K original images as queries. For each query, we retrieved the top-57 most similar images and tallied how many of the 57 near-replicas were missing. We computed the average missing rate for each near-replica category (colorizing, modulation, cropping, rotation, scaling, downsampling, shading, blurring, framing, changing format, and changing contrast). Figures 2 (a) and (b) show the training results. (Since the training results include many curves, we display them in two figures for clarity.) Figures 2 (a) and (b) show that the optimal range of m is between 110 and 130, where the missing rates are the lowest.



(a) Using color and Wavelet texture features

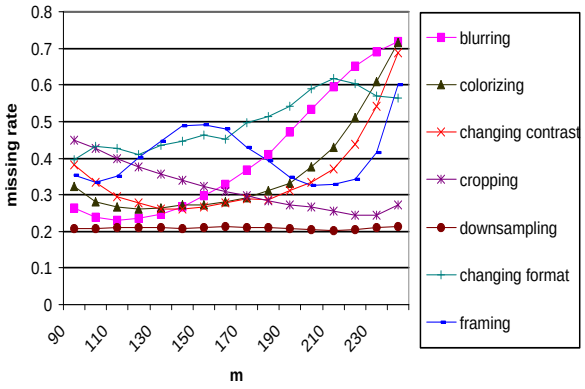


(b) Using color and Gabor texture features

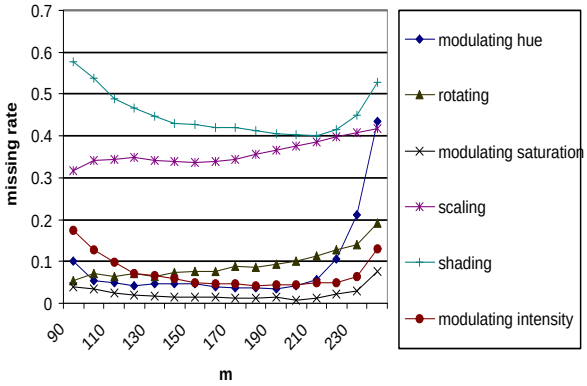


(c) Using color, Wavelet and Gabor texture features

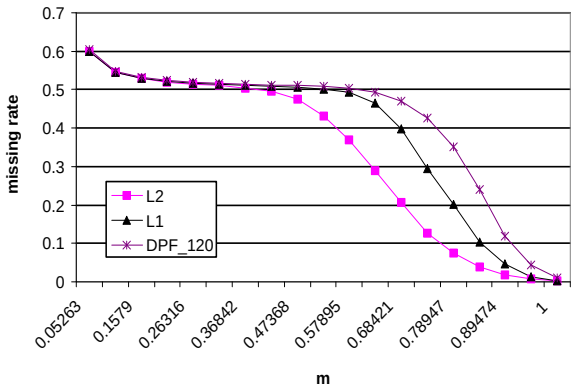
Figure 1. Predicting optimal m values for different feature combinations



(a) Predicting optimal m

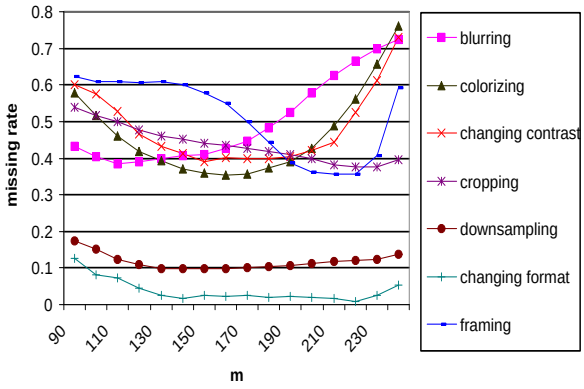


(b) Predicting optimal m

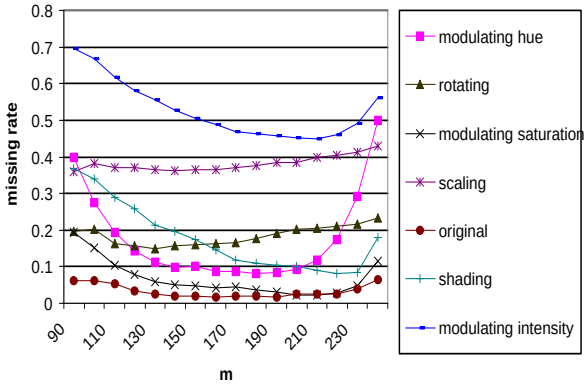


(c) Precision/Recall

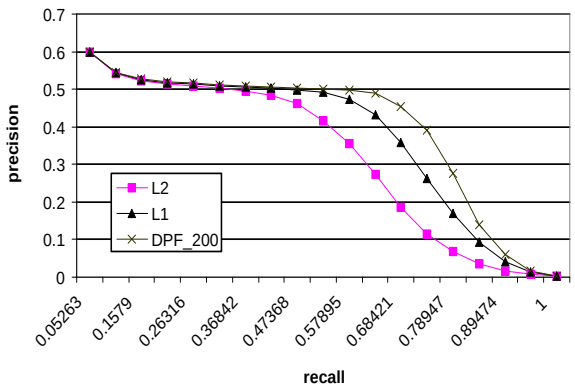
Figure 2. Using original images as queries



(a) Predicting optimal m

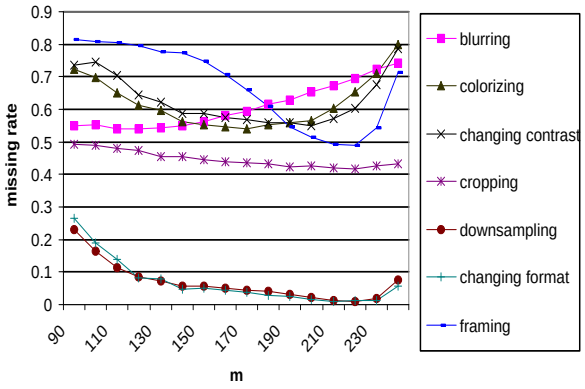


(b) Predicting optimal m

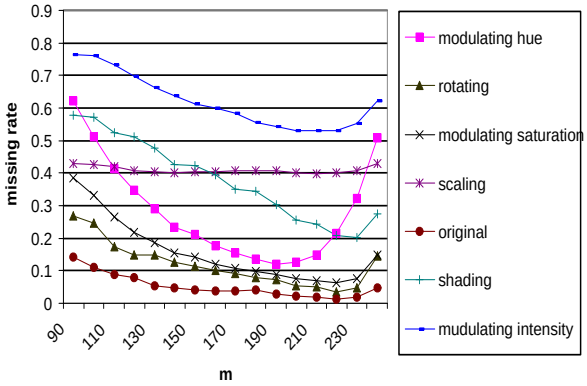


(c) Precision/Recall

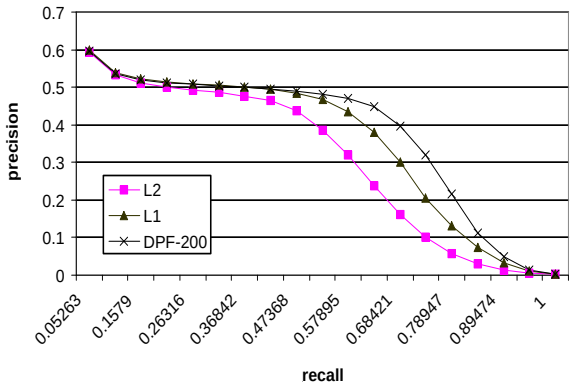
Figure 3. Using 10% cropped images as queries



(a) Predicting optimal m

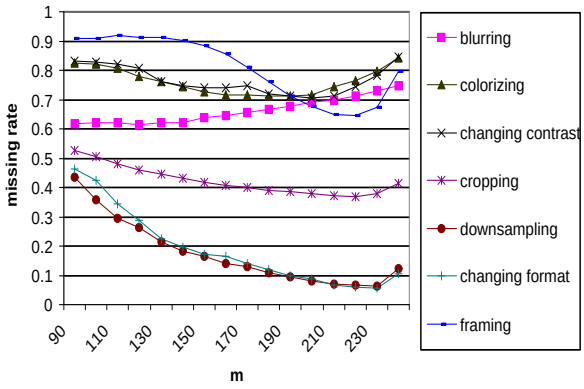


(b) Predicting optimal m

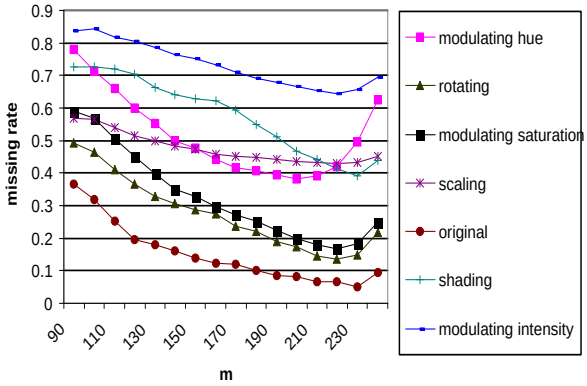


(c) Precision/Recall

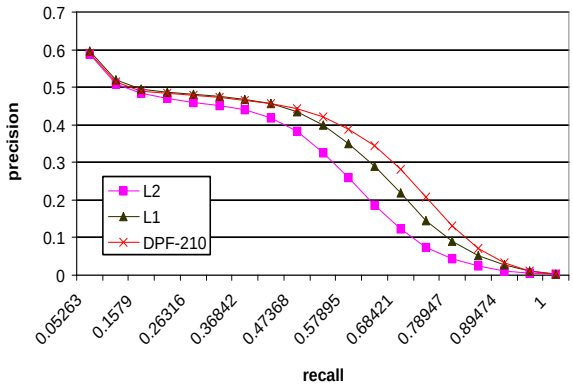
Figure 4. Using 20% cropped images as queries



(a) Predicting optimal m

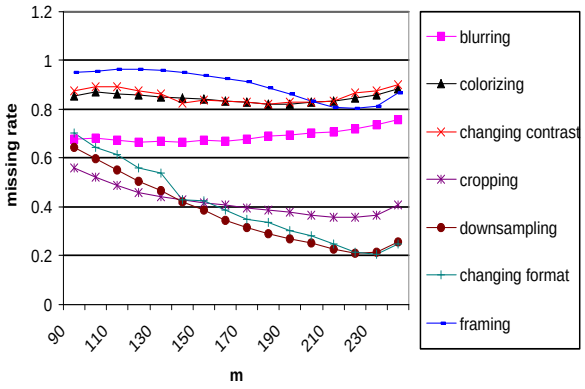


(b) Predicting optimal m

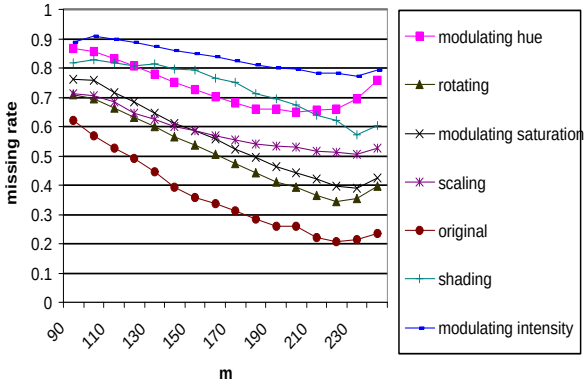


(c) Precision/Recall

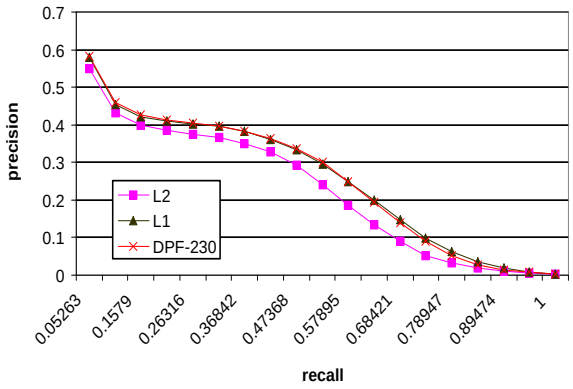
Figure 5. Using 30% cropped images as queries



(a) Predicting optimal m

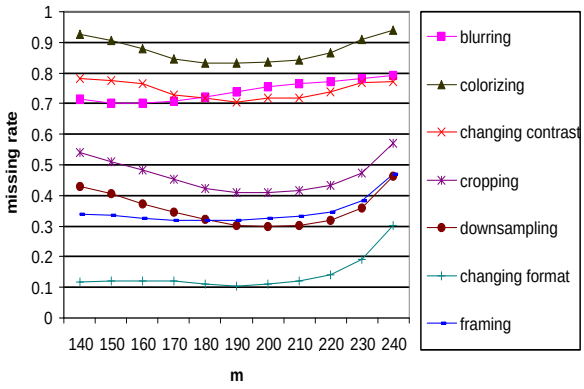


(b) Predicting optimal m

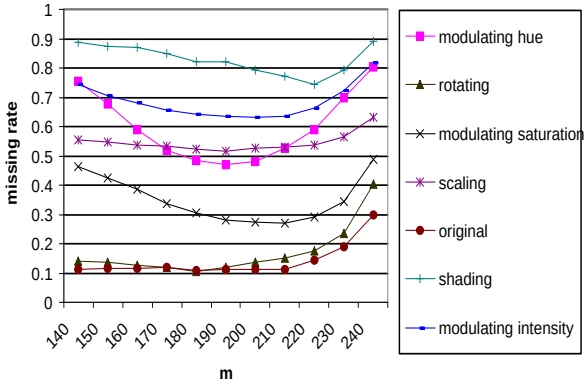


(c) Precision/Recall

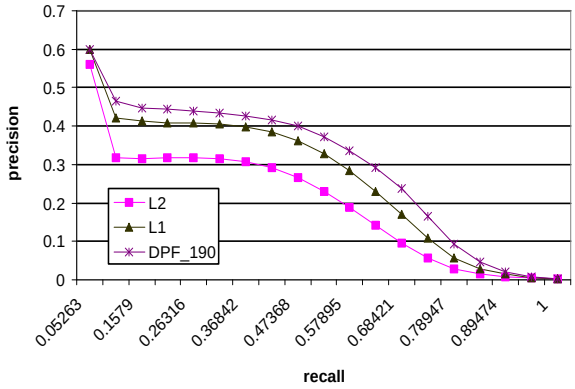
Figure 6. Using 40% cropped images as queries



(a) Predicting optimal m



(b) Predicting optimal m



(c) Precision/Recall

Figure 7. Using framed images as queries

Testing the effectiveness of DPF

We compared DPF with L1 and Euclidean distance functions using the testing dataset. After setting $m = 120$ based on the training results, we used the original images in the testing dataset to perform queries and recorded the average precision and recall of finding the near-replicas. The testing result (Figure 2 (c)) shows that DPF works significantly better than both L1 and Euclidean distance functions. For instance, when the recall rate is 68.421% (about two out of three near-replicas are found), the precisions of Euclidean, L1, and DPF distance functions are 27.9%, 46.3%, and 49.4%, respectively.

4.2.2. Using Transformed Images as Queries

We also used transformed images as queries to test the robustness of DPF. We first used cropped images to perform queries. Figures 3 (a), (b), and (c) show the training and testing results of using 10% cropped images. Figures 4 (a), (b), and (c) display the results of using 20% cropped images. Figure 5 (a), (b), and (c) present the results of using 30% cropped images. Figures 6 (a), (b), and (c) show the results of using 40% cropped images to perform queries. DPF consistently outperforms the other two distance functions in all cases.

Finally, we used framed image to perform queries. Figures 7 (a), (b), and (c) show that when recall is 68.421%, the precisions of Euclidean, L1, and DPF are 14.3%, 22.8%, and 28.4%, respectively.

4.3. Observations

From the experimental results, we make two observations. First, all testing results show that DPF achieves higher accuracy than both Euclidean and L1 distance functions. Second, DPF can easily capture near-replicas produced by changing format, downsampling, rotating, and minor cropping. Though DPF outperforms the other functions, it is not as effective in finding e.g., severely cropped near-replicas. By examining the images, we believe some severely cropped images do not really exhibit perceptual similarity to the original, and therefore, missing detecting these near-replicas is understandable and perhaps acceptable. For instance, Figure 8 presents seven near-replicas of a bear image (the original is at the top-left corner). The images that DPF misses are displayed in the second row, and they do not look similar to the original image.



(a) Original (b) 10% Downsampling (c) 10% Cropping (d) Framing



(e) Scale down (10 folds) then up (f) Colorizing (30%/70%/90%) (g) 50% Cropping (h) 60% Cropping

Figure 8. Image samples

5. CONCLUSION

To tackle the image copy-detection problem, we have employed a robust set of features to represent images and used DPF to measure similarity. Extensive experimental results show that DPF outperforms traditional distance functions by significant margins.

We plan to enhance our work in two directions. First, we will build a much larger database to test the robustness of our feature set. Second, we currently obtain an "on average optimal" m value through training. Ideally, however, the optimal m should be determined in a pair-wise fashion, i.e., each pair of images should have their m setting. We plan to improve DPF by finding m in an adaptive way.

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